

VIGNESHWARI JAYAPRAKASH

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EDUCATION

Master of Science in Data Science (Computing and Decision Analytics)

May 2026

Arizona State University, Tempe, AZ

GPA: 4.00/4.00

- Coursework: Machine Learning, Data Mining, Distributed Database Systems, Big Data, Optimization

Bachelor of Information Technology

Anna University, India

GPA: 3.6/4.00

- Coursework: Data Structures and Algorithms, Object Oriented Programming, Probability & Statistics

WORK EXPERIENCE

Graduate Intern

Jun 2025 – Dec 2025

State of New Mexico - Office of Cybersecurity

USA

- Built an XGBoost-based compliance classification model for state water and wastewater vendors, achieving 90%+ accuracy in identifying non-compliant suppliers and using SHAP explainability to support contract review decisions
- Developed Power BI dashboards tracking 1M+ infrastructure devices to surface real-time system health KPIs for executive leadership, earning direct recognition from the CIO
- Built an AI chatbot microservice for cybersecurity operations using LangChain, OpenAI APIs, and ChromaDB over 100K+ infrastructure log records, reducing engineering triage time by 70% (45 minutes to 3 minutes)
- Engineered SQL ETL workflows, exploratory data analysis (EDA) and feature engineering pipelines extracting 14 operational features from infrastructure logs, improving machine learning F1-scores by 22% (0.71 to 0.87)

Technology Analyst (Specialized in Data Science)

Oct 2017 - Jan 2019

Infosys Limited (Client: BNSF Railway, USA)

India

- Developed a Logistic Regression fraud detection model to classify fraudulent transactions, improving detection accuracy on 1M+ daily transactions and contributing to an estimated \$1.5M+ in annual fraud savings
- Built predictive maintenance models for 5,000+ locomotive assets using XGBoost and ARIMA, eliminating \$400K/year in unplanned maintenance costs
- Led the migration of legacy mainframe reporting systems to Tableau dashboards and automated ETL workflows, reducing reporting time by 50% and operational costs by 70%

Senior Software Engineer

Oct 2013 - Sep 2017

Infosys Limited (Client: BNSF Railway, USA)

India

- Conducted A/B testing on the Employee Medical Testing system to evaluate and optimize feature enhancements, leading to a 30% improvement in system efficiency and user satisfaction
- Designed a K-Means clustering framework for vendor segmentation across 50K+ suppliers, reducing manual evaluation cycles by 30% and saving 600+ analyst hours per quarter
- Trained over 5 junior analysts in advanced data visualization methodologies, improving team performance

ACADEMIC PROJECTS

SentinelRAG: Compliance Screening System [\[Github\]](#)

- Built a production RAG platform using LangChain, Ollama, ChromaDB, and FastAPI that achieved 99% recall and 100% precision across 18,700+ compliance entities, replacing ~400 manual reviews per week

E-Commerce Hybrid Recommender [\[Github\]](#)

- Improved recommendation relevance and scalability across 2M+ products by deploying a Two-Tower retrieval architecture with FAISS ANN indexing and a LightGBM ranking pipeline, increasing MRR by 22%, NDCG@10 by 18% and reducing inference latency by 60% (180ms → 72ms)

Agentic Supply Chain Demand Orchestrator [\[Github\]](#)

- Increased supply chain decision-making efficiency by 35% by developing a forecasting and RAG-based analytics platform using XGBoost, ARIMA, LangChain and ChromaDB, achieving <8% MAPE across 46M+ records

SKILLS

Core: Python (Pandas, NumPy, Scikit-learn, Statsmodels), R, SQL

Machine Learning: Feature Engineering, Model Evaluation (Precision, Recall, F1-Score, ROC-AUC), Supervised Learning (Logistic Regression, Random Forest, XGBoost, Gradient Boosting), Unsupervised Learning (K-Means, Hierarchical Clustering)

Generative AI: LLMs, RAG, LangChain, Prompt Engineering, Vector Databases, OpenAI API, Anthropic API, Ollama

Experimentation & Statistics: A/B Testing, Hypothesis Testing, Multivariate Testing

Cloud & Data Engineering: AWS (S3, Glue, Athena, SageMaker), Azure, Databricks, dbt, Snowflake

Visualization: Power BI, Tableau, Looker